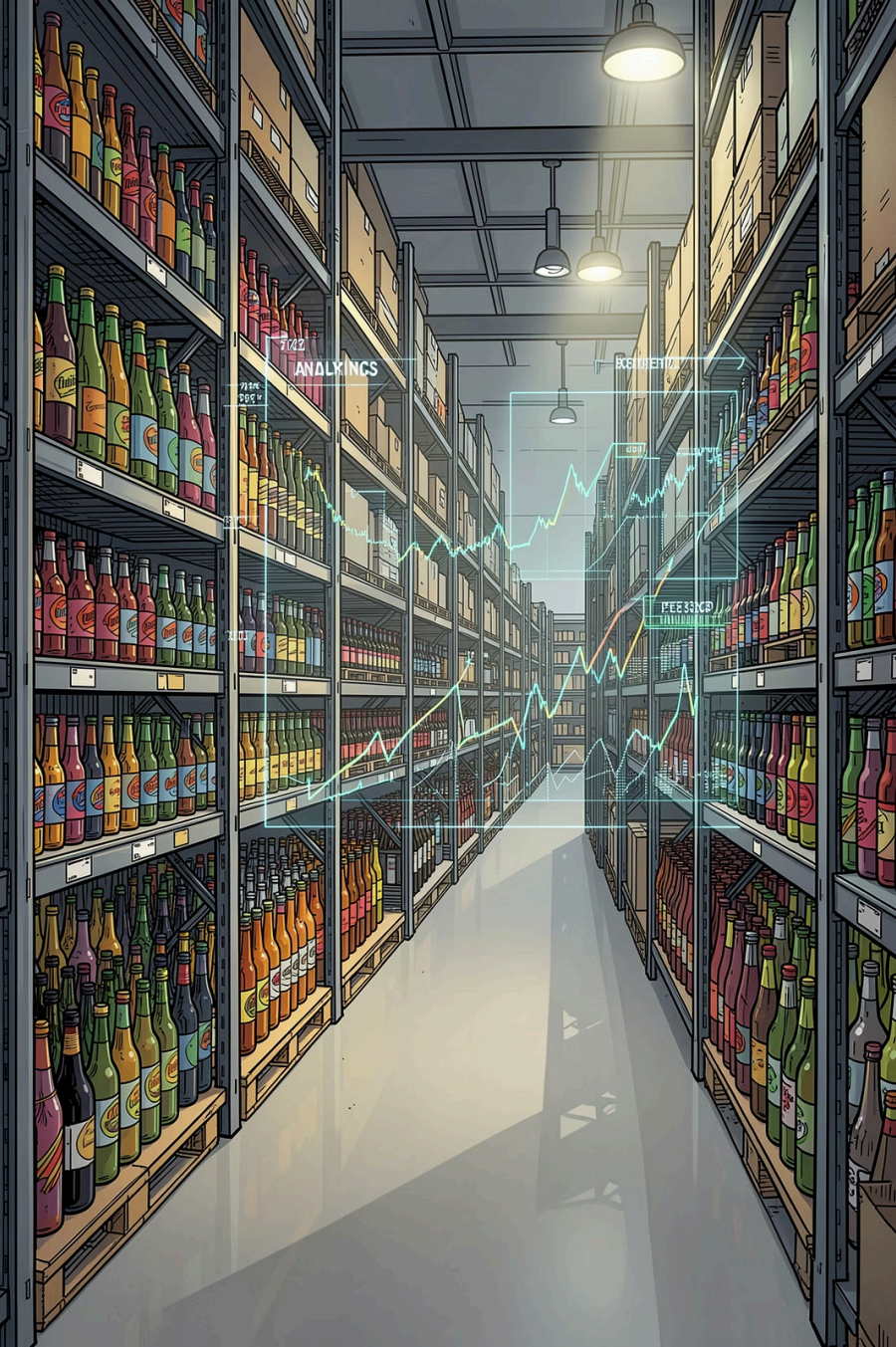




Vendor Performance Analysis

Retail & Wholesale Distribution — Analyzing vendor performance across sales, profitability, inventory efficiency, and pricing strategy to drive actionable business outcomes.

Business Problem

Underperforming Brands

Identify brands needing promotional or pricing adjustments to improve sales.

Top Vendor Identification

Determine which vendors contribute most to sales and gross profit.

Bulk Purchasing Impact

Analyze how order size affects unit costs and margins.

Inventory Efficiency

Assess turnover rates to reduce holding costs and free up capital.



Dataset Overview

Six CSV files totaling ~**15.6 million rows**, linked by **VendorNumber** and **Brand** keys.

Table	Rows	Description
sales	12.8M	Sales transactions by vendor & brand
purchases	2.37M	Purchase orders from vendors
end_inventory	224K	Closing inventory snapshot
begin_inventory	207K	Opening inventory snapshot
purchase_prices	12K	Product pricing reference
vendor_invoice	5.5K	Freight and invoice data

Tools & Workflow



Data Ingestion

Load CSVs into SQL Server via Python & SQLAlchemy



Cleaning & Merging

Handle NULLs, duplicates; build master table with SQL CTEs



Exploratory Analysis

Summary stats, distributions, correlations (Pandas, Seaborn)



Statistical Testing

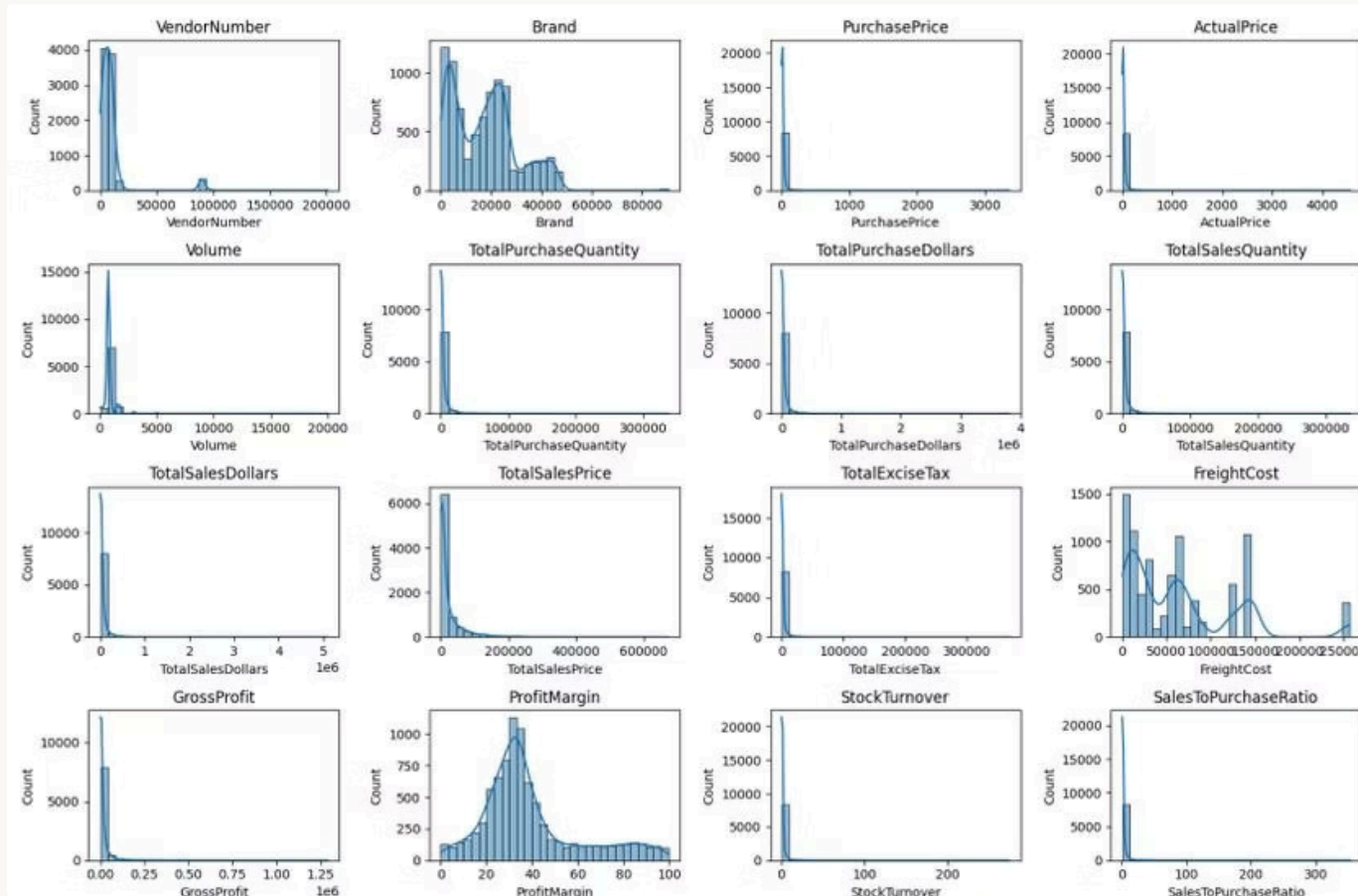
T-tests, confidence intervals (SciPy, NumPy)



Dashboard & Report

Interactive Power BI dashboard with slicers; documentation on GitHub

Key EDA Findings



Negative Gross Profit

Min -\$52K — some products selling at a loss

Zero Sales Items

Products purchased but never sold
— slow-moving stock

Price & Freight Variation

Purchase price: \$0.36–\$5,682;
Freight: \$0.09–\$257K

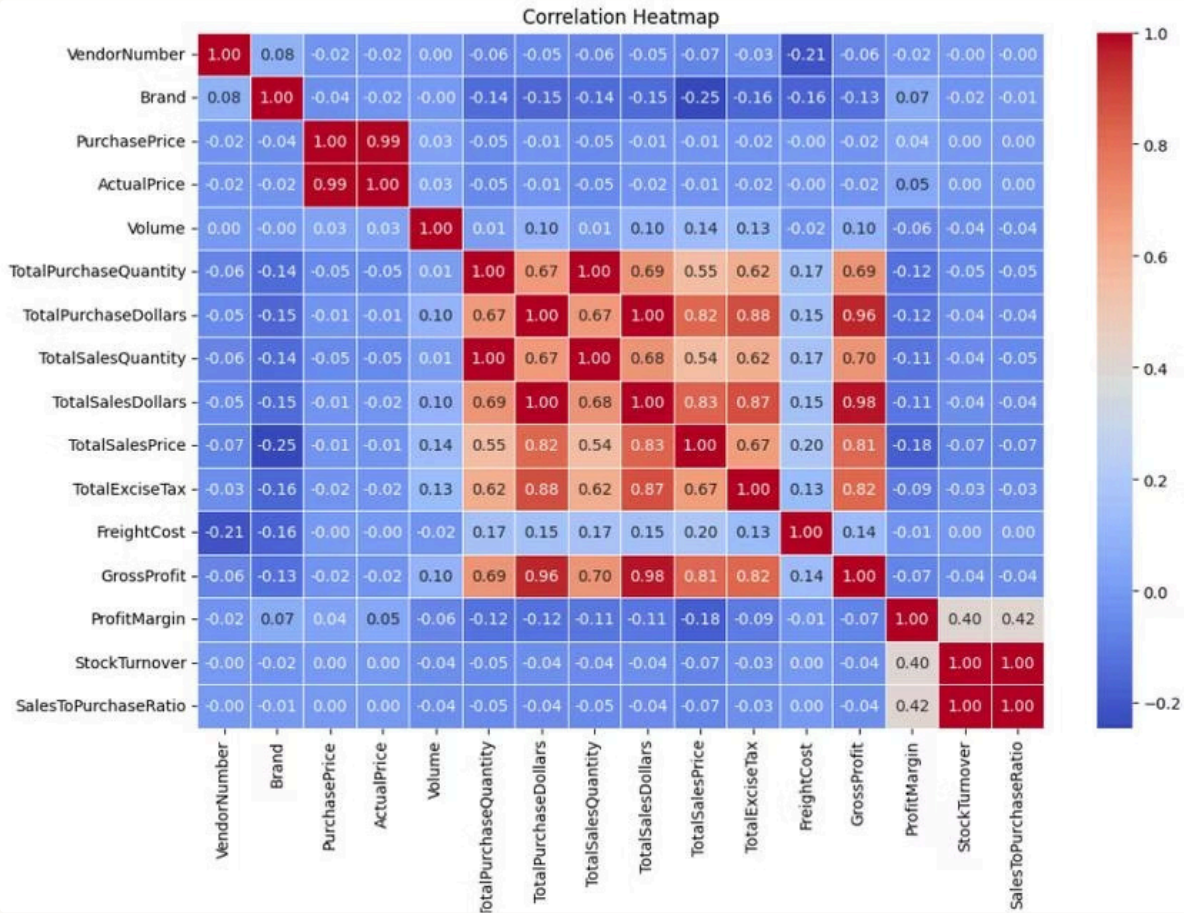
Stock Turnover Range

0 to 274.5 — some products sell fast, others sit idle

Correlation & Top Performers

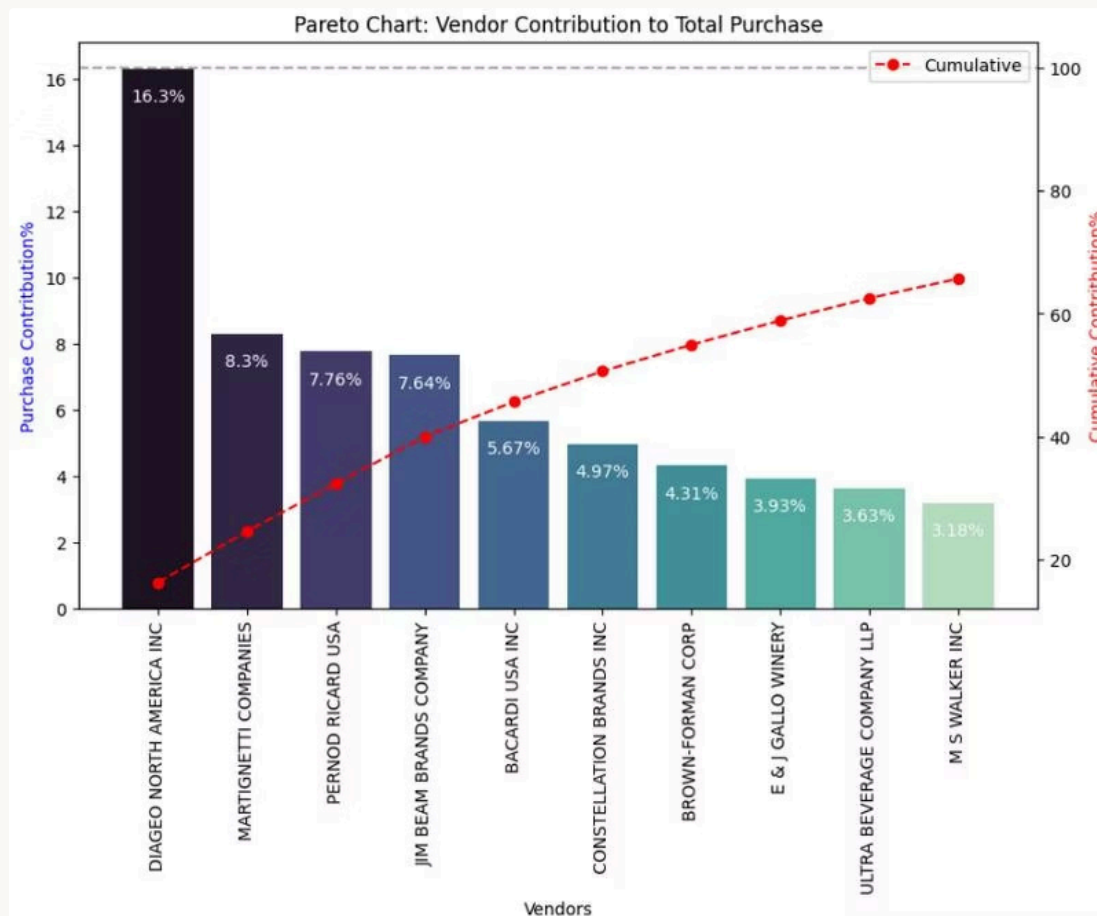
Key Correlations

- Purchase Qty vs Sales Qty: **1.00** — efficient inventory turnover
- Stock Turnover vs Profit Margin: **0.40** — faster inventory = higher margins
- Profit Margin vs Sales Price: **-0.18** — higher prices may compress margins



Top vendors by sales: Diageo North America (\$67.99M), Martignetti Companies (\$39.33M), Pernod Ricard USA (\$32.06M).
Top brands: Jack Daniels (\$7.86M), Tito's Handmade Vodka (\$7.40M), Grey Goose (\$7.21M).

Pareto Analysis & Bulk Purchasing



Vendor Concentration

Top 10 vendors contribute **65.69%** of total purchase dollars. Diageo alone accounts for 16.3%.

Bulk Discount Effect

Large orders achieve **~72% cost reduction** vs. small orders — average unit price drops to just \$10.78 at scale.

Inventory Efficiency & Statistical Testing

Low Turnover Vendors (Turnover < 1)

- Diageo North America: **\$722K** unsold inventory
- Jim Beam Brands: **\$555K** unsold inventory
- Pernod Ricard USA: **\$471K** unsold inventory
- Total unsold capital: **\$2.71M**

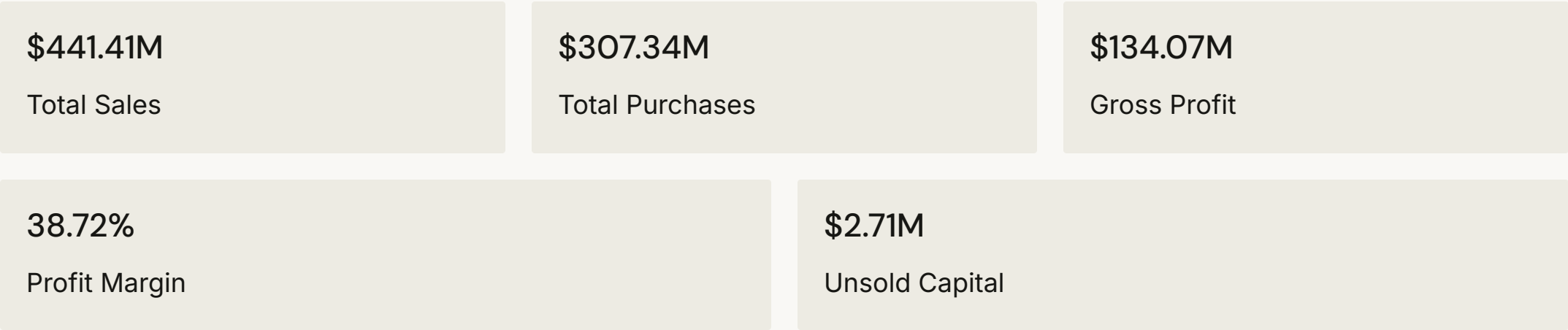
95% Confidence Intervals — Profit Margin

- Low-Performing Vendors: **40.48% – 42.62%** (mean ~41.5%)
- Top-Performing Vendors: **30.74% – 31.61%** (mean ~31.2%)

 **T-Test:** $T = -17.64$, $p = 0.0000 \rightarrow$ Significant difference confirmed. Low-sales vendors maintain higher margins (premium pricing); high-sales vendors use volume strategy.



Power BI Dashboard



Interactive slicers enable dynamic filtering by vendor, brand, and category. Cross-filtering connects Pareto charts, sales rankings, and statistical comparisons in one view.

Recommendations

1 **High-Performers (High Sales, Lower Margins)**
Test 2–5% price increases; negotiate freight; bundle products to lift gross profit per transaction.

2 **Low-Performers (Low Sales, Higher Margins)**
Promote high-margin products via targeted campaigns; expand distribution; offer volume discounts.

3 **Inventory Management**
Review turnover <1 items for markdowns or discontinuation; reduce \$2.71M unsold capital via demand forecasting; incentivize bulk purchasing.

4 **Vendor & Promotional Strategy**
Strengthen top-10 vendor relationships (65.69% of purchases); target low-sales/high-margin brands for promotions without eroding profitability.

